
TinyAudit: On-Device, Uncertainty-Aware Fairness Audits that Survive (and Expose) Compression

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Abstract

Machine learning is moving onto microcontroller-class hardware, where models are aggressively compressed (pruned, quantized) to fit. Responsible-AI audits, in contrast, are almost always run on full-precision models on a workstation, and they typically look at fairness, uncertainty, and explainability in isolation. We present TinyAudit, an auditing pipeline that measures all three together on the compressed artifact rather than the original, and that profiles its own cost. Theory already tells us calibration and parity cannot generally agree, and that compression harms underrepresented groups; we claim neither as new. Our contribution is measuring what those results do to a deployed model, reported as means over 10 seeds. First, the two lenses select different groups: the ordering by selection rate and the ordering by per-group calibration error cross on UCI Adult and again on COMPAS, whose favorable direction is inverted. Second, this compression effect is protocol-dependent: one-shot pruning drives an MLP’s demographic-parity difference toward zero while per-group calibration error climbs, at every sparsity, but standard prune-then-fine-tuning removes that divergence through 70% sparsity and leaves only a residual at 90%, where accuracy is already lost below the majority-class rate. Either way a parity-only audit cannot separate a healthy model from a collapsed one, so calibration has to be measured on the compressed artifact. Third, the uncertainty view is not redundant with calibration: predictive entropy ranks the groups differently from per-group calibration error (mean $|\rho| = 0.21$ over the race cells), so both are needed. Finally, the audit’s own peak working set is a fixed floor of about 0.6 MB, plus roughly 2.4 KB per sample once batches are large, and constant in test-set size when streamed. That constant does not yet fit the 32 to 512 KB parts at the low end of the TinyML range, and we report the gap rather than claim it closed.

1 Introduction

TinyML runs inference on microcontrollers with 32 to 512 KB of SRAM [19]. To fit that budget, deployed models are pruned and quantized. The auditing tools that would check those models for fairness were built for a different world: cloud notebooks, full-precision weights, and demographic labels available at test time [14, 2, 8]. Existing TinyML benchmarks measure latency, accuracy, and energy, but not fairness or explainability [19]. The result is a gap. The models most likely to be compressed are the least likely to be audited, and when they are audited, the audit sees a model that is not the one deployed.

A second problem is that fairness audits usually stop at point predictions. A model can look fair on selection rates while its uncertainty is biased against a group [13], and abstaining on high-uncertainty inputs does not by itself repair group imbalance [17]. Uncertainty quantification now fits on KB-

37 sized devices [7], but no existing system evaluates those estimates through a fairness lens, and no
38 fairness toolkit fits on the device.

39 TinyAudit closes both gaps at once: a seven-stage pipeline that measures fairness, uncertainty, and
40 explainability on the compressed model, inside a memory envelope sized for the device being au-
41 dited.

42 **Relation to known results.** Both results have theoretical ancestors and we want to be precise about
43 what is new. That calibration and parity-style criteria cannot generally hold at once is established
44 [11, 16], in Chouldechova’s case on COMPAS specifically [4]; that compression disproportionately
45 damages underrepresented groups is likewise documented [9, 3]. We claim neither as new. What is
46 new is the setting: those results come from full-precision models and workstation tooling, and the
47 model deployed to a microcontroller is neither. The two interact in a way that matters operationally:
48 under one-shot compression, parity and calibration move in *opposite* directions, so the metric most
49 audits report improves as the model gets worse; standard fine-tuning confines this to aggressive
50 sparsity, but either way no toolkit measures calibration on the compressed artifact inside the device’s
51 own budget.

52 Contributions:

- 53 1. An on-device audit instrument: a seven-stage pipeline running fairness, uncertainty, and ex-
54 plainability on the compressed model, emitting a one-page traffic-light card and a machine-
55 readable manifest.
- 56 2. A measurement showing the calibration/parity tension is group-selective in practice on the
57 deployed model: the two lenses rank sensitive groups differently on Adult and replicate on
58 COMPAS, whose favorable direction is inverted (Section 4).
- 59 3. A protocol-dependent failure mode for compression-era audits: under one-shot pruning,
60 demographic parity collapses while per-group calibration error rises at every sparsity; stan-
61 dard prune-then-fine-tuning confines this to aggressive (90%) sparsity, but in both regimes
62 a parity-only audit cannot distinguish a healthy model from a collapsed one (Section 5).
- 63 4. Evidence that the uncertainty lens is not redundant with calibration: predictive entropy
64 ranks the sensitive groups differently from per-group ECE (mean $|\rho| = 0.21$ over the race
65 cells), so “uncertainty-aware” earns its name rather than restating calibration (Section 6).
- 66 5. A feasibility measurement of the audit itself, which to our knowledge has not been reported
67 before: peak working set is a fixed floor of about 0.6 MB plus roughly 2.4 KB per sample
68 at large batches, and is constant in test-set size when streamed (Section 7).

69 2 The TinyAudit instrument

70 TinyAudit takes a fitted model and a labeled test set and produces a one-page audit card plus a JSON
71 manifest that every stage writes to. The card is generated from the manifest and never hand-edited;
72 every figure in it is backed by a committed CSV. The stages, in order:

73 **1. Profile.** Parameter count, an analytic FLOPs proxy, peak RAM (via `tracemalloc`), and wall-
74 clock per sample. **2. Point fairness.** Demographic-parity difference (DP), equalized-odds difference
75 (EO), and disparate-impact ratio (DI), each with an explicitly pinned favorable direction and a valid
76 range enforced by property-based tests (DP, EO $\in [0, 1]$; DI a non-negative ratio). **3. Uncertainty.**
77 MC dropout (MLP only), a lightweight perturbation ensemble, and a QUTE-style early-exit estima-
78 tor [7]; each returns predictive entropy, variance, and mutual information. **4. Uncertainty-aware**
79 **fairness.** Per-group predictive entropy, per-group expected calibration error (ECE), and a selective-
80 fairness AUC that integrates parity as low-confidence predictions are progressively abstained (com-
81 puted and emitted to the manifest; not analyzed in this paper). **5. Explainability.** A dependency-free
82 occlusion explainer reporting per-group feature importance and which features enter one group’s top-
83 k but not another’s. SHAP and LIME wrappers ship with the package but stay out of the audit path,
84 being costly and fragile at the edge [18, 15]. The stage is counted in the footprint of Section 7; it
85 yields no fairness result here (Section 8). **6. Compression.** int8 dynamic quantization or magnitude
86 pruning, applied *before* every metric above, so the audit sees the deployed model. **7. Render.** The
87 one-page, traffic-light-coded card.

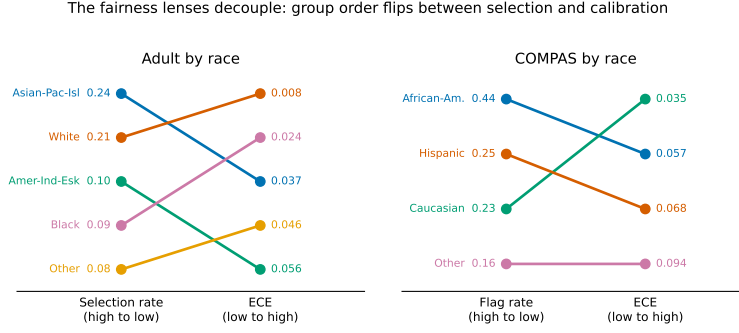


Figure 1: Finding 1. Rank slope charts for Adult (left) and COMPAS (right) by race, uncompressed logistic model, means over 10 seeds. Each line joins a group’s rank by selection/flag rate to its rank by per-group ECE. Crossing lines mean the two lenses order the groups differently.

88 The audit’s working set is dominated by per-sample buffers, so it streams in small batches at a peak
 89 RAM independent of test-set size (Section 7). The whole public API is one call:

```
90 from tinyaudit import audit
91 card = audit(model=clf, data=(X_test, y_test), sensitive="sex",
92             methods=["fairness", "uncertainty", "xai"], compression="int8")
93 card.to_pdf("adult_lr_int8.pdf")
```

94 3 Experimental setup

95 **Datasets.** UCI Adult [12] (36,631/12,211 train/test, 101 features), COMPAS [1] (4,629/1,543, 5
 96 features; here the positive outcome is being flagged high-risk, so higher is worse), and Folktables
 97 ACSIncome [5] (146,748/48,917, 8 features, 2018 California). Sensitive attributes are sex and race
 98 throughout. The first two are the near-universal fairness benchmarks [6]; ACSIncome is the modern
 99 replacement for Adult’s 1994 distribution [5].

100 **Models.** Scikit-learn estimators on standardized features: logistic regression, an MLP with two
 101 hidden layers of 32 units, and a decision tree. Scaling matters, since without it the Adult MLP sits
 102 near the majority-class rate (0.54 accuracy) versus 0.84 with it. The scaler is fit on train only and
 103 applied outside the model, so compression and perturbation reach the raw weights.

104 **Reproducibility.** A single seed flag seeds random, NumPy, torch, and PYTHONHASHSEED. Every
 105 headline number is a mean $\pm$ std over seeds 0 to 9; the prune-then-fine-tune arm (Section 5) retrains
 106 the surviving weights for six short masked rounds over those same seeds. Determinism is guaranteed
 107 on Linux/x86; elsewhere we expect last-digit float drift, which the std columns absorb. All models
 108 land in published ranges (Adult logistic regression 0.852 ± 0.003 accuracy at ROC-AUC $0.906 \pm$
 109 0.002 , Adult MLP 0.837 ± 0.003 , COMPAS logistic 0.680 ± 0.006 , ACSIncome MLP 0.808 ± 0.003 ;
 110 the full grid is a committed CSV). The fairness results therefore sit on models that work, not on
 111 degenerate predictors.

112 4 Result 1: the two lenses select different groups

113 Point-prediction fairness asks how often each group receives the favorable outcome. Calibration
 114 fairness asks how honest the model’s confidence is per group. The impossibility results [11, 4] say
 115 these cannot generally agree; what they do not say is *which* group each lens singles out on a given
 116 deployed model, which is what an auditor has to act on. On the uncompressed logistic model the
 117 two lenses rank the sensitive groups differently, and the two orderings cross rather than shift together
 118 (Figure 1).

119 **Adult by race** (five groups). By selection rate Asian-Pacific-Islander (0.240) and White (0.207)
 120 rank highest and the smaller groups lowest; calibration reorders them. White is only second-most-
 121 selected yet by far the best calibrated (ECE 0.008 ± 0.002 , three to seven times lower than every

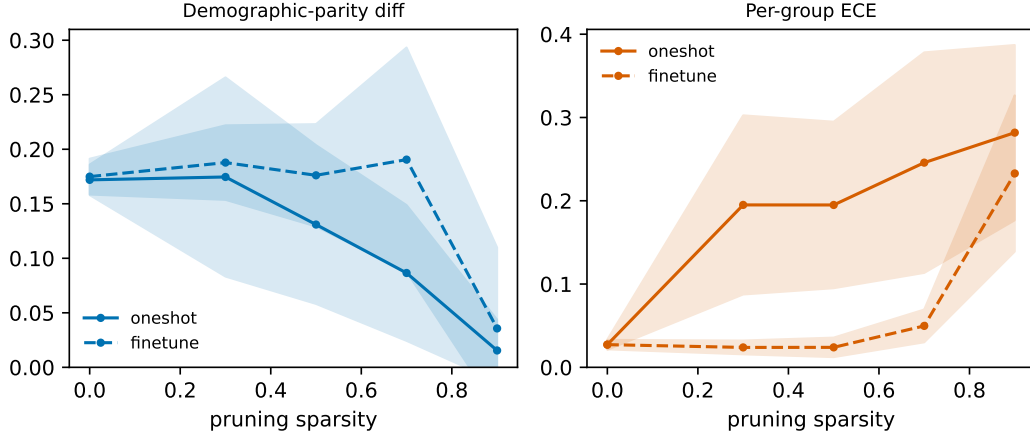


Figure 2: Finding 2. Adult MLP by sex under magnitude pruning, one-shot (solid) vs prune-then-fine-tune (dashed); bands are ± 1 std over 10 seeds. One-shot pruning drives demographic parity down and per-group ECE up at every sparsity; fine-tuning holds both flat through 70% and lets only a residual reappear at 90%, where accuracy can no longer be recovered.

other group), while American-Indian-Eskimo is worst (0.056 ± 0.019). The smaller groups carry higher, noisier calibration error that the selection view does not surface. **By sex** the disagreement inverts: a large selection gap (Male 0.253 vs Female 0.079) with negligible calibration difference (ECE 0.009 vs 0.014).

COMPAS by race (positive outcome = flagged high-risk, so higher is worse; groups with $n < 10$ dropped). African-American defendants are flagged at roughly twice the rate of the next group (0.441 ± 0.015 vs Hispanic 0.249 ± 0.046), the well-known point disparity. Yet the worst-calibrated group, Other (ECE 0.094 ± 0.029), is flagged *least* (0.164 ± 0.033), and the best calibrated is Caucasian (0.035 ± 0.016). The reordering therefore holds on a second dataset whose favorable direction is inverted, so it is not an Adult-specific label-convention artifact.

Why it matters on-device. Sensitive labels are rarely available at inference time, so on-device audit triggers lean on confidence scores as a proxy [10]; if the lenses disagree about which group is worst served, a uniform confidence threshold can track the wrong disparity [17, 13]. Reporting both is the cheap fix, and TinyAudit does so by construction.

5 Result 2: one-shot compression hides unfairness; fine-tuning confines it

We prune the Adult MLP to increasing sparsity and re-run the full audit at each level under two protocols that differ in exactly one step: one-shot magnitude pruning, and pruning followed by masked fine-tuning, in which the surviving weights are retrained while the pruned weights are held at zero — the standard deployment practice (Figure 2).

One-shot pruning produces the divergence at every level. By sex, demographic parity falls from 0.172 ± 0.014 to 0.015 ± 0.028 at 90% while per-group ECE climbs from 0.027 ± 0.006 to 0.282 ± 0.105 ; by race, DP $0.152 \rightarrow 0.022$ and ECE $0.050 \rightarrow 0.314$. But the model degrades at every rung: accuracy falls to 0.50 at 90%, below the 0.76 majority-class rate. It looks fairer on parity precisely as it stops predicting.

Fine-tuning removes the divergence through 70% sparsity. Retraining the survivors keeps accuracy at 0.77–0.84 through 70% sparsity, and with it demographic parity stays put (sex ≈ 0.18 , race ≈ 0.16) and per-group ECE stays low (0.02–0.07). Up to aggressive sparsity, the one-shot divergence is an artifact of skipping the fine-tuning step.

A residual survives only at 90%, where the model breaks anyway. There fine-tuning cannot restore the model (accuracy 0.59 ± 0.24 , still below the majority rate) and a weakened divergence returns (sex DP 0.036, ECE 0.233; race DP 0.036, ECE 0.262). Seed-paired against one-shot at 90%, fine-tuning significantly lowers the calibration damage (ECE $\Delta = -0.05$, 95% CI $[-0.08, -0.02]$)

154 but does not significantly change the parity collapse ($\Delta = +0.02$, CI crosses zero). The COMPAS
 155 logistic model at 90% one-shot pruning shows the same endpoint in the extreme: it predicts one
 156 class for everyone, giving $DP = 0$ and $DI = 1$, perfect parity by construction.

157 The upshot for auditors is protocol-independent: a parity-only audit cannot tell a healthy fine-tuned
 158 model from a collapsed one — both can read $DP \approx 0$ — so calibration has to be measured on the
 159 compressed artifact, whichever protocol shipped it.

160 **6 Result 3: the uncertainty lens is not the calibration lens**

161 We call the audit uncertainty-aware, so it is fair to ask whether the uncertainty view adds anything
 162 the calibration view does not already give: per-group ECE also reads confidence, so is “uncertainty-
 163 aware” just “calibration-aware” renamed? We test it directly. From the same perturbation ensemble
 164 we take the pure uncertainty quantities it already produces (per-group predictive entropy, ensemble
 165 variance, and mutual information) and ask whether they single out a *different* group than calibration
 166 does (Figure 3, Appendix A).

167 They do. Over 10 seeds the mean absolute Spearman correlation between the entropy ordering
 168 and the ECE ordering is 0.21 across the three race cells (Adult logistic -0.40 , Adult MLP $+0.18$,
 169 COMPAS logistic -0.06), so the two lenses do not rank the groups together. Entropy is not noise: on
 170 Adult it tracks the *selection-rate* ordering (Spearman $\approx +0.75$), so it forms its own axis rather than
 171 restating calibration. On sex (two groups, so the contrast is a disparity gap rather than a correlation)
 172 the entropy gap between groups is large where the ECE gap is negligible (Adult 0.154 vs 0.006;
 173 Adult MLP 0.155 vs 0.023). We also report the one cell where the two agree (COMPAS by sex,
 174 where the ECE gap is the larger of the two). The uncertainty view therefore adds information that a
 175 calibration-only audit would miss. The one-shot Result 2 collapse also replicates on ACSIncome, a
 176 third and larger Census benchmark (Appendix B).

177 **7 Feasibility: what the audit itself costs**

178 Audit tooling is normally not profiled at all: the model’s footprint is reported and the auditor’s is
 179 assumed free because it runs on a workstation. If the audit is to run where the model runs, that
 180 assumption has to be checked. We profile the full three-method audit with `tracemalloc` and report
 181 the pipeline’s own peak RAM and wall-clock, not the audited model’s (Table 1, Appendix C).

182 The working set is dominated by per-sample buffers: the feature matrix, the ensemble’s stacked
 183 prediction arrays, and the explainer’s perturbation batch. Two regimes appear across the Adult
 184 logistic sweep (Table 1): a fixed floor of about 0.6 MB that dominates at small batches (610 KB at
 185 batch 64, 641 KB at batch 256, essentially flat) and a per-sample term of about 2.4 KB that dominates
 186 once the batch is large (2.48 MB at 1024, 9.83 MB at 4096, a slope through a near-zero intercept). A
 187 single additive fit overstates the middle of the range by nearly a factor of two, so the honest summary
 188 is the larger of the two terms:

$$\text{peak RAM} \approx \max(0.6 \text{ MB}, 2.4 \text{ KB} \times \text{batch size}), \quad (1)$$

189 the two crossing near a batch of 256. The structural claim is straightforward: an arbitrarily large
 190 test set streams through the audit in fixed-size batches at constant peak RAM, so audit cost is set
 191 by the batch size the engineer picks, not by how much test data exists. Against the standard toolkit
 192 the in-house metric path is not the bottleneck: TinyAudit’s point-fairness stage returns the identical
 193 demographic-parity value as Fairlearn (its reference oracle) while using $8\text{--}11\times$ less peak memory
 194 (e.g. 164 vs 1,352 KB at a 4096-row batch on Adult, 47 vs 514 KB on COMPAS).

195 The absolute constant needs more care. At a 256-row batch the audit peaks at 0.61 to 0.95 MB in
 196 under 0.1 s: not within the 32 to 512 KB parts at the low end of the TinyML range, but within reach of
 197 the high end, such as Cortex-M7 parts with roughly 1 MB of SRAM. The engineer controls only the
 198 per-sample term, not the 0.6 MB fixed cost, so driving that fixed cost below 512 KB is the concrete
 199 target this measurement identifies. Two caveats: the audited Adult MLP itself peaks at 6.0 MB, so
 200 it stands in for an MCU-class model rather than being one (the 62 KB COMPAS logistic model
 201 is the realistic case), and `tracemalloc` excludes the interpreter while counting Python overhead a
 202 compiled port would avoid. The scaling law is robust; the absolute constant is not.

8 Limitations and conclusion

Features are standardized before fitting, so the numbers describe the scaled-input regime. The uncertainty ensemble is five lightly-jittered copies of the compressed model, not five retrains, so uncertainty magnitudes are relative. The prune-then-fine-tune arm covers the MLP; the logistic and ACSIncome compression results are reported under one-shot pruning only, and whether fine-tuning likewise confines their collapse is left to future work. int8 models’ float weights are unreachable, so their uncertainty columns are blank; decision trees have no dense weights and sit out the compression grid. Footprint numbers are measured on x86, not on a microcontroller. The explainability stage yields no fairness result: the top- k feature divergence between groups is non-monotone under pruning, within seed noise, and does not replicate on COMPAS, so we report that stage as instrument and cost, not evidence. Neither result we do report is theoretically novel [11, 4, 9]; the contribution is measuring them on the model that ships, where those effects prove measurable, group-selective, and — under one-shot pruning at any sparsity, and even after fine-tuning at aggressive sparsity — actively misleading, at under a megabyte per streaming batch: a parity-only audit cannot separate a healthy compressed model from a collapsed one.

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A Uncertainty-vs-calibration rank charts

Figure 3 shows the per-cell rank slope charts underlying Result 3. The main-text numbers (mean $|\rho| = 0.21$ across the race cells; the per-cell Spearman values; the sex disparity contrast) are self-contained; this figure is the visual form of that finding.

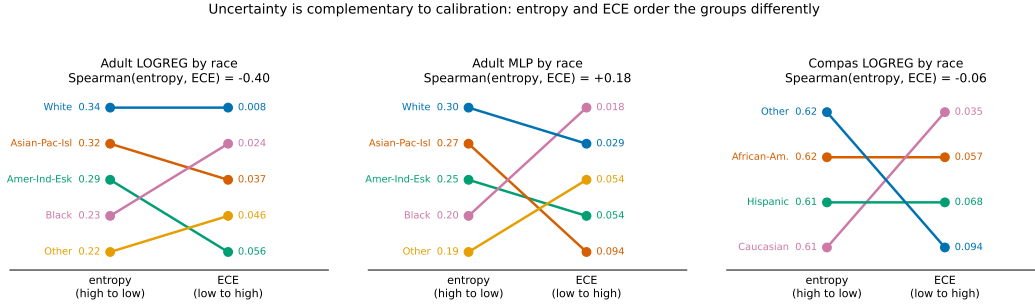


Figure 3: Rank slope charts by race for the three framing-critical cells, uncompressed, means over 10 seeds. Each line joins a group’s rank by predictive entropy to its rank by per-group ECE; crossing lines mean the uncertainty and calibration lenses order the groups differently. Panel titles give the per-cell Spearman rank correlation.

B ACSIncome replication of the compression collapse

Under one-shot pruning, the Result 2 effect is not specific to Adult and COMPAS. On ACSIncome (logistic, three seeds, test set subsampled to 8,000 rows for cost), pruning to 90% sparsity drives the demographic-parity difference by sex from 0.034 to 0.000 while accuracy collapses from 0.781 to 0.589 and the positive-prediction rate falls to zero — the “fairest” model in the sweep is again the most broken — and per-group ECE disparity rises from 0.008 to 0.112. The same silent-collapse mechanism appears on a modern, larger Census benchmark.

C Audit pipeline footprint

Table 1: Audit pipeline footprint (the pipeline itself, not the audited model), profiled with tracemalloc over 3 seeds. Peak RAM is governed by the streaming batch size, not the test-set size.

Dataset	Model	Batch	Audit peak RAM (KB)	Audit wall-clock (s)
Adult	logreg	64	610	0.03
Adult	logreg	256	641	0.06
Adult	logreg	1024	2,477	0.11
Adult	logreg	4096	9,826	1.10
Adult	MLP	256	946	0.08
Adult	MLP	4096	9,824	1.28
COMPAS	logreg	256	609	0.01
COMPAS	MLP	1543	1,181	0.05